Neuro-Symbolic Causal Reasoning for Cautious Self-Adaptation under Distribution Shifts

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***Abstract*—[Context] Learning-enabled self-adaptive systems (SAS) derive their behavior (policy) from taking actions on non-stationary environments whose distribution shifts can violate safety-critical constraints that are risky and costly to mitigate via ad hoc policy updates.**

[State-of-the-Art] Most mitigating approaches lie in the spectrum of relearning a new policy from scratch to transferring certain policy components (areas of state-action space) *as-is* and relearning others. While the former approaches suffer from *catastrophic forgetting*, the latter are impacted by *negative transfer*. Because these phenomena threaten safety-critical constraints for SAS, one needs to reason about trade-offs (cost-risk-benefit) between the two ends of the spectrum. However, that is not possible with current solutions due to their over-reliance on calibrated thresholds to identify the affected policy components.

[Contribution] To remedy this, we study representations for reasoning about environmental behavior and how they influence transfer choices. We combine three representations: (1) a *symbolic abduction tree* to trim the decision-space of transfer choices, (2) a *causal model* of the environment to inform the decision on transfer choices, and (3) a *deep reinforcement learning model* to maintain a policy for the learning-enabled SAS. We integrate these representations into a *neuro-symbolic causal reasoning framework* to support learning-enabled SAS subject to non-stationary environments.

[Results] We evaluate our framework for an adaptive traffic signal control (TSC) application, where constant changes in street conditions induce the learned policy to violate safety and performance constraints. Two sets of results show that the framework is capable of (1) informing transfer choices that allow the policy to recover from violated constraints and (2) updating the policy with transfer, thus restoring the safety constraint while showing larger cumulative rewards than learning from scratch or picking a similar threshold-based policy.

***Index Terms*—Causal Inference, Reinforcement Learning, Self-Adaptive Systems, Neuro-Symbolic**

I. Introduction

Context Learning-enabled safety-critical systems [57], like autonomous vehicles and intelligent traffic signal control (TSC), derive their behavior, e.g., Deep Reinforcement Learning (DRL) policy, by taking actions within environments that are subject to different types of *distributions shifts*,[1]i.e when the data a model sees during deployment differs from the data it was trained on, which if left unmitigated [26], [41], can cause mispredictions of adaptation actions, which can violate the system's performance and safety constraints.

State-of-the-Art Gap Most mitigation approaches fall into a set of adaptation tactics [11] that can be as simple as replacing a pre-trained model or re-learning the model from data (existing or relabeled), or more involved transfer between models (reuse). Initial approaches triggered model training with the arrival of new datasets [13], [23], [28] or after detection of distribution shift in dataset stream [34]. Because of their reliance on the data, these models are more vulnerable to *catastrophic forgetting*. Catastrophic Forgetting happens when a model trained on a given task loses its prediction capability after being trained on a second task [43]. Various approaches were studied to mitigate catastrophic forgetting [37]. *Negative Transfer* happens when, after knowledge transfer, the new knowledge from one task negatively impacts the performance of another task [69]. To mitigate this, new adaptive approaches identify the aspects of the model that require update at distinct levels of granularity, for instance, a new classification task [25], a reduced feature space [53], or a new system component [12]. Because these approaches reuse knowledge from previous models, they are then more vulnerable to negative transfer. In either case, there is the risk of discarding knowledge.

We argue in this paper that decisions about how to update the models must go beyond correctly getting the timing of the shift (trigger) [12], [41], its impact (magnitude of the mispredictions) [12], [26], and the viable tactics to update the prediction model (relearn, unlearn, transfer) [11], [53]. One needs to take into account how the distribution shifts in the environment are causally associated with the adaptation options and their effects, i.e., the decision structure. Support for this type of reasoning is overlooked in current learning-enabled self-adaptation approaches.

Research Problem To remedy this, we investigate reasoning about transfer as a trade-off between knowledge acquired via formal specifications and knowledge learned from data (training a DRL policy). We make a first step in this investigation by posing two research questions: (RQ.1) Is there a trade-off between constraint specification and choices of knowledge transfer? (RQ.2) Is the choice of knowledge transfer more effective than the two extremes of training from scratch (zero transfer) or replacing with a pre-trained policy (total transfer)?

Approach The need to reason about specified versus learned knowledge suggests the combination of the two AI traditions captured now under the umbrella of *Neuro-Symbolic* (NeSy) methods [3], [7], [24]: neural networks, e.g., DRL, and symbolic reasoning based on explicit knowledge representation using formal languages including *formal logic* and causality. In

[1]Covariate shifts, prior distribution shifts, and concept drifts [21], [48].

this paper, we extend our neuro-symbolic vision for principled transfer learning in autonomic systems [4]. Our approach consists of a framework for Neuro-Symbolic (NeSy) causal reasoning that combines three types of representations: (1) a *Symbolic Abduction Tree* (SAT) to trim the state-space of possible adaptations, (2) a *causal graph* to capture interdependencies and estimate the impact (look-ahead effect) of an adaptation, and (3) a *deep reinforcement learning model* to derive a policy that minimizes the training cost and maximizes the cumulative reward while satisfying the safety constraints for *cautious*, hence, safer system adaptation.

Contribution (1) a NeSy framework that enables causal reasoning for learning-enabled safety-critical systems, (2) a systematic evaluation of the trade-offs in knowledge constraint specification versus learning from data based on an adaptive TSC simulator (SUMO) [2], and (3) a demonstration of the advantage of partial transfer versus the extremes of zero (learn from scratch) versus simply replacing with a pre-trained model. Finally, the source code, data sets, and DRL methods used in this paper are publicly available [35].

Tour Section II presents the preliminaries. The NeSy causal reasoning approach is presented in Section III and evaluated in Section IV. Section V discusses the state-of-the-art and Section VI concludes the paper.

II. Preliminaries

A. Causal Inference & Discovery

Causal inference under Pearl's formulation [51] relies on a Bayesian network, which is a directed acyclic graph G (DAG), with nodes V representing covariates and their marginal probabilities or expectations and edges E representing the causal associations whose strengths (effects) are measured in conditional probabilities. The aforementioned DAG is also referred to as *Causal Graph* G. *Causal discovery* is the process of discovering the structure of G whereby one relies on various causal assumptions[2] and applies conditional independence tests to eliminate possible edges empirically [51]. Concerning inference, one can also generate from G a set of equations, called *Structural Causal Model* (SCM), where each covariate corresponds to a node in G and participates in a regression model with coefficients representing effects of these covariates on the outcome. Given an SCM and Pearl's *Backdoor criteria* [51], we can determine the control covariates, i.e., *adjustment set*, which guarantee, under causal assumptions (direction of correlations), that intervening on a covariate X (setting its value) produces an effect on an outcome $o_i = \beta X_i + \alpha M$, where M is the set of mediators and confounders. The equations are generated by blocking paths or confounders, for instance, by applying Pearl's Backdoor criteria [51] to identify the variables to be controlled for, i.e., whose influence on the effect must be blocked because of undesirable paths (confounders) or indirect effects (mediators). Hence, except for the intervention variable I_i, all other variables of the right-hand side of the equation operate as control variables [31], i.e., fixing the value of one or more covariates, hence allowing for estimating the effect on one or more covariates (outcome variables). These causal effects can be *direct* or *indirect*, i.e., mediated by one or more covariates.

Intervention I_i while leaving all other covariates intact. Note that when covariates are on the right-hand side of the equation, they work as control variables, which have the role of blocking the influence of these covariates on the intervention effect. In this paper, we use Bayesian regression to fit the SCM and to estimate the posterior distribution of the intervention effects. Typical choices of interventions are independent variables, i.e., nodes in G without parents.

A causal Path Ψ conveys the effect of one intervention covariate X_i to one outcome variable o_i, which can represent a direct (De), indirect (Ie), or total (Te) effect. through a set of mediating covariates M. These effects are estimated by sampling posterior distributions from SCM equations. While the paths Ψ are generated by traversing the graph, the effects are obtained from the SCM.

[2]*Causal Sufficiency* (no hidden confounders), *Faithfulness* (independence relations in the data are caused only by the underlying structure of the graph that generated the data), *Causal Markov Condition* (any node V is independent of all its non-descendants given the set of all its parents) [52].

B. Deep Reinforcement Learning (DRL)

Reinforcement Learning (RL) is a machine learning method where an agent learns to make decisions through trial and error. This problem is often modeled mathematically as a Markov Decision Process (MDP). A finite MDP is a tuple (S, A, π, R, λ) where S is the *state-space*, i.e., a representation of the environment at a given time and the *action-space* is captured as A, which are the choices available to the agent. The *reward* function R is the immediate reward of executing an action $a \in A$. A *Policy* π is a function that maps each state $s \in S$ to a probability distribution over the set of possible actions and is achieved through training the DRL agent. $\lambda \in [0, 1]$ is the discount factor. The DRL agent is trained to make a sequence of decisions that maximize cumulative rewards over time [61]. DRL algorithms incorporate deep learning to solve RL problems with high-dimensional MDP states where the policy is represented by a neural network.

C. Abductive Reasoning

Abductive reasoning, or *abduction*, is the process of inferring the most likely explanation or cause c for a given observation o [5]. Abduction differs from *deduction*, i.e., deriving consequences from premises, and *induction*, i.e., generalizing from specific instances, by focusing on generating hypotheses to explain an observation. *Causal inference* constitutes one of the main areas where abductive reasoning is used to generate causal explanations, connecting causes and their effects—see [39]. Given an observation $o \in \mathbb{O}$, the abduction process generates hypotheses for explanation $c \in \mathbb{C}$ such that $c \rightarrow o$, where, from an occurrence of c and the rule $c \rightarrow o$, suggests an occurrence of c as a plausible explanation for o. The rules used for abduction are typically specified using material implication that are implicitly, i.e., by their use, interpreted as causal relationships—see [50].

Logic-based approaches are one of the several formal models used to specify the abductive process [50]. First-Order Logic (FOL), with its ability to express relationships between objects and entities, can be used to model the domain knowledge, including facts, rules, and constraints that connect hypotheses to observations in abductive reasoning. The domain knowledge is represented as theory $\mathcal{T}$ in a logical first-order language $\mathcal{L}$. The logical specification of the abduction knowledge makes it amenable to *symbolic reasoning* approaches. Symbolic approaches rely on the explicit representation of knowledge using formal languages, and FOL provides a formal foundation for representing and manipulating knowledge using symbols, logical rules, and inference mechanisms.

An *abduction problem* for causal relationships according to [50] is formulated as $(\mathcal{C}, \mathcal{O}, \mathcal{T})$ defined over the first-order language $\mathcal{L}$, with $\mathcal{C}$ a set of causes, $\mathcal{O}$ a set of observations (outcomes), and $\mathcal{T}$ a theory representing the domain knowledge defined in $\mathcal{L}$. If a sentence A in $\mathcal{L}$ is found as the result of an abductive process in searching for an explanation of $\mathcal{O}$, A must satisfy the following conditions: (1) *consistency*: A is consistent with $\mathcal{T}$, i.e., the theory and the explanation should not contradict each other; (2) *entailment*: $\mathcal{O}$ follows from A and $\mathcal{T}$, i.e., $\mathcal{T} \cup A \models \mathcal{O}$; (3) *parsimony*: A is *subset-minimal*, i.e., it avoids the choice of a set of sentences as explanations that contain a proper subset which itself constitutes a valid explanation. Abduction is then the process that picks out some member of A with the highest *explanatory power* [50]. Explanatory power EP is a function of the ratio between the conditional probability of a hypothesis H given the evidence E and the likelihood of the evidence, i.e., $EP = F(\frac{P(H|E)}{P(E)})$. EP lies between zero and one, where one means that the evidence is fully explained [58].

D. Adaptive Traffic Signal Control

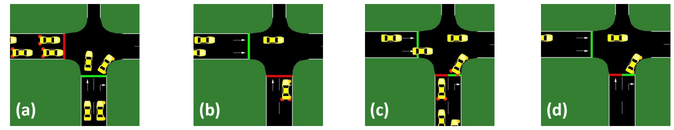
Simulation of Urban MObility (SUMO) [2] is an open-source, microscopic, and continuous road traffic simulation framework designed to handle large-scale networks. It is widely used for various Intelligent Traffic Management System (ITMS) applications, e.g., online traffic monitoring systems [42] and intelligent signal control strategies [45]. In this paper, SUMO is employed to simulate the ITMS environment for developing an adaptive TSC system. We use a reinforcement agent, henceforth DRL agent, controlling a traffic light that has an unprotected right turn condition reproduced in SUMO under the following conditions:[3]

1) $N \Rightarrow S$ *Green*, $W \Rightarrow E$ *Red* (Figure 1-a).
2) $W \Rightarrow E$ *Green*, $N \Rightarrow S$ *Red* (Figure 1-b).
3) Unsafe unprotected right[4] $S \Rightarrow E$ (Figure 1-c).
4) Safe protected right $S \Rightarrow E$ (Figure 1-d).

The covariates regulating the environment are friction (Fr), emergency brake (EB), average speed (S), waiting time (WT), collisions (Co), and desired speed (DS). The goal is

[3] $W \Rightarrow E$ (West to East), $N \Rightarrow S$ (North to South), *Green* means open to enter the traffic cross, *Red* means closed to cross.

[4] The unprotected right turn can be safe (no risk of collisions) or unsafe (with risk of collisions).



South $\Longrightarrow$ North (*Green*) West $\Longrightarrow$ East (*Red*) | West $\Longrightarrow$ East (*Green*) South $\Longrightarrow$ North (*Red*) | Unsafe unprotected right turn (collisions) | Safe unprotected right turn (no collisions)

Fig. 1: SUMO scenarios

to optimize for both traffic flow (minimize WT) and safety (minimize Co).

III. Approach

The cautious adaptation approach relies on a NeSy causal reasoning process combining three types of representations: (1) a *Symbolic Abduction Tree* (SAT) to trim the state-space of possible adaptations, (2) a *causal graph* G to capture interdependencies and estimate the impact (look-ahead effect) of an adaptation, and (3) a *deep reinforcement learning model* (DRL) to derive a policy that minimizes the cost of training and maximizes the cumulative reward while satisfying the safety constraints during the adaptation. We use out-of-the-box methods for the causal discovery of G and DRL, rendering the proposed framework extensible for any causal discovery and DRL technique. We first show how to construct an SAT for logic-based abductive reasoning in Section III-A, then we present an overview of the NeSy causal reasoning adaptation integrated into the activities of a MAPE-K feedback loop in Section III-B. The inner workings of the causal reasoning engine are discussed in Section III-C.

A. Symbolic Abduction Tree

We propose a *symbolic abduction tree* (SAT) as a logical and hierarchical structure to represent and organize the abductive reasoning process. For a given abduction problem $(\mathcal{C}, \mathcal{O}, \mathcal{T})$, SAT allows for systematic exploration of possible explanations that, according to the theory $\mathcal{T}$, could account for a set of observations—see Section II-C. The root of the tree represents the observation or phenomenon to be explained, i.e., shift in distribution. The nodes represent intermediate explanations; the branches represent logical inferences connecting observations to potential explanations; and finally, the leaves represent accepted explanations or situations where no further elaboration is possible. In the context of causal reasoning, a path in SAT ends at a leaf node and represents a sequence of causal explanations linking an observation, i.e., outcome, to potential root causes through abductive reasoning. Terminal nodes represent primitive causes that cannot be further broken down. Edges represent causal relationships, i.e., the causal effects between potential causes and outcomes.

1) SAT generation: Obtained via causal discovery, the causal graph G is used to generate a symbolic abduction tree (SAT). A discovered causal graph capturing the relationships for the covariates in SUMO is shown in Figure 2. For each *outcome* node in G, i.e., a node with an out-degree of zero, an SAT with the outcome node as its root is generated. Each root node in SAT corresponds to one of the system/ DRL agent's *(safety) constraint* Cs— a constraint Cs is an objective that the agent has to satisfy while managing a self-adaptive system,

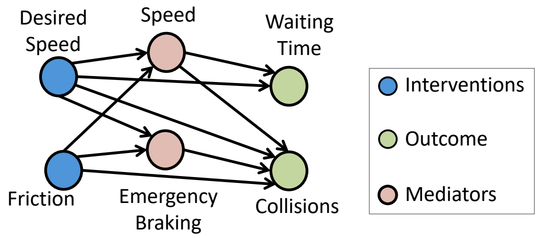


Fig. 2: Causal graph for TSC agent generated (discovered) from traces

e.g., minimum performance level or a maximum safety risk. In this paper, we are concerned with safety constraints. As G is a DAG, the corresponding abduction trees could be achieved via reverse traversal and Breadth-First Search (BFS) starting from outcome nodes in G. The result is an SAT for each outcome node in G. Figure 3 shows the two possible SATs, one for each outcome node in the causal graph described in Figure 2—see Figure 4 for a flowchart describing the generation of SAT based on a causal graph G.

The relationships between the outcome (observation) and potential causes in SAT are in a first-order logical structure and are captured in a first-order language $\mathcal{L}$. The theory $\mathcal{T}$ is a set of first-order logic sentences (e.g., axioms, rules, facts) that capture the general domain knowledge. SAT enables symbolic abductive reasoning to select the most appropriate intervention to mitigate the violation of a constraint Cs. Each internal node V in SAT corresponds to mediators in the corresponding caudal graph G. The leaf nodes of SAT correspond to the DRL agent's actions that are instantiated as interventions on G.

2) Reasoning with SAT: For a given SAT where its root captures an observation, i.e., a distribution shift in one of the outcome variables, the abduction process entails checking the possible explanations, i.e., all the children nodes in SAT, against the theory $\mathcal{T}$. Each possible explanation is assessed regarding its satisfaction of the consistency, entailment, and parsimony conditions described in Section II-C.

We use an exemplary scenario in the context of a TSC agent in SUMO to describe the reasoning via SAT. **Scenario**: there has been a collision although the emergency brakes were applied (observation $\mathcal{O}$). We want to abduce possible causes (explanations) $\mathcal{C}$ from a traffic knowledge base (theory $\mathcal{T}$) and $\mathcal{O}$. We define the first-order predicates $old(x)$: recent value of x; $\Delta(x) : x - old(x)$; $\rho(x, \beta) : \frac{\Delta x}{old(x)} \leq \beta$; $EB(x)$: x emergency brakes. For the abduction problem $(\mathcal{C}, \mathcal{O}, \mathcal{T})$, let $\mathcal{T}$ be a theory consisting of the following propositions:

(1) $\forall x \neg(\Delta(speed(x)) > 0 \wedge \Delta(friction(x)) < 0)$
(2) $\forall x \neg(\Delta(speed(x)) = 0 \wedge \Delta(friction(x)) < 0)$
(3) $\forall x \neg(\Delta(speed(x)) < 0 \wedge \Delta(friction(x)) > 0)$
(4) $\forall x (\Delta(speed(x)) < 0 \leftarrow \Delta(friction(x)) < 0)$
(5) $\forall x (collision(x) \leftarrow \neg EB(x))$
(6) $\forall x (collision(x) \leftarrow \Delta(speed(x)) > 0$
(7) $\forall x (\rho(collision(x), 5\%))$
(8) $\forall x, \exists b (\rho(collision(x), 2\%) \leftarrow (\Delta(speed(x)) = 0 \wedge \Delta(friction(x)) = b \wedge b < 0))$

Abductive reasoning: for the given observations $\mathcal{O} = \{collision(a), EB(a)\}$, a possible explanation $c \in \mathcal{C}$ could be $c = \{\Delta(speed(a)) > 0) \wedge \Delta(friction(a)) > 0\}$. Note that a is a constant value for the variable x. The predicate $EB(a) \in \mathcal{O}$ rules out proposition (5) in $\mathcal{T}$, thus, $collision(a) \in \mathcal{O}$ suggests that proposition (6) holds. As a result, $\Delta(speed(a)) > 0$ is true. Proposition (1) implies that if there is an increase in $speed$, i.e., $\Delta(speed(a)) > 0$, for the proposition to hold, $friction$ should increase as well, i.e., $\Delta(friction(a)) > 0$. Because both predicates should be true, the clause $c = \{\Delta(speed(a)) > 0) \wedge \Delta(friction(a)) > 0\}$ is the *best* possible explanation for $\mathcal{O}$. $\mathcal{C}$ is the best explanation because according to the consistency condition (see Section II-C) c is consistent with $\mathcal{T}$, the theory and the explanation entail the observation: $\mathcal{T} \cup c \models \mathcal{O}$, and c is subset-minimal, i.e., there is no subset of $c \in \mathcal{C}$ that explains $\mathcal{O}$. The theory $\mathcal{T}$ may capture simple (7) and complex (8) constraints whereby proposition (7) constrains the increase in *collision* to a maximum of 5% and proposition (8) indicates that in cases where *speed* does not change, the reduction in *friction* is only allowed as long as *collision* does not increase more than 2%.

The rules captured via theory $\mathcal{T}$ concern safety propositions that have two objectives: (i) identify the constraint violation conditions, e.g., proposition (7,8) and (ii) restrict the types of causal interventions that can be extracted from the SAT, e.g., not allow simultaneous increase of speed and reduction of friction—proposition (1).

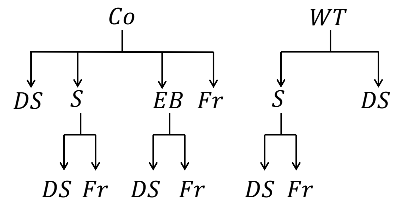


Fig. 3: Generated SATs from causal graph with Collision Co (left) and Waiting Time WT (right) as roots

B. Interleaving NeSy Causal Reasoning into MAPE-K loop

In the proposed framework, the learning and reasoning are interleaved into the distinct activities of the MAPE-K as in Figure 5—see Algorithm 1 for summarized steps.

Knowledge comprises three representations: a *symbolic abduction tree* SAT (Section III-A), *causal model* G (Section II-A), and *DRL policy* π (Section II-B). They constitute the knowledge base $\mathcal{K}$ for both the reasoning and learning steps. These representations are *generated/discovered* once during the *Initialization* and, then, are continuously *updated* during the execution of the adaptation loop.

1) Initialization: The three representations are generated prior to the very first execution of the adaptation loop, i.e., during initialization and before the DRL agent is able to learn to adapt. The steps are executed in the order they are presented:

Initial policy generation - the agent is equipped with an arbitrary initial DRL policy π_0.

Causal discovery - the DRL agent is deployed with π_0 on a set of M *environment configurations* $\bar{\Phi}_{1:M}$—environment configuration $\bar{\Phi}_{1:N}$ is a vector of N attributes representing

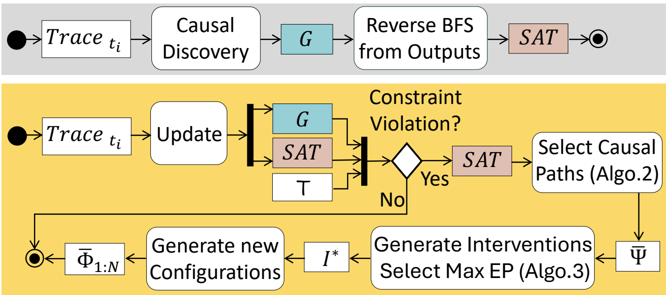


Fig. 4: Interplay between SAT and G during Initialization (gray, $i = 0$) and Causal Reasoning Engine (yellow, $i > 0$)

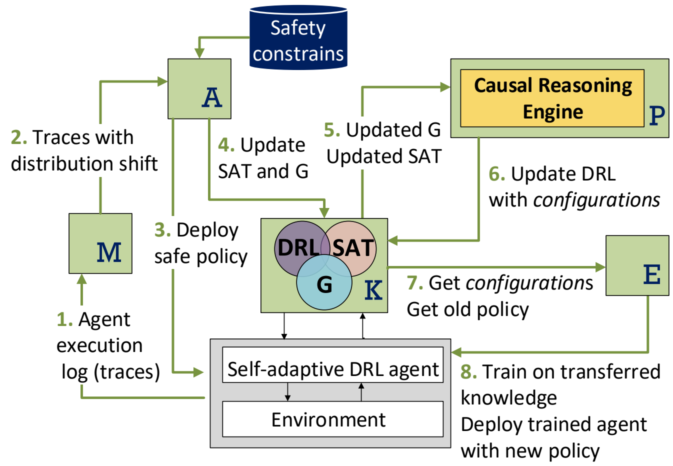


Fig. 5: Overview of the NeSy causal approach integrated into a MAPE-K adaptation loop

the covariates that can be subject to interventions. Typically, covariates are nodes in the causal graph G without parents.

The DRL agent execution traces are used to discover the structure of the causal graph G. We chose the Peter & Clarke algorithms [63], which rely on testing for conditional independence to discover causal association. It is considered an efficient and popular algorithm, hence its choice favors the reproducibility of the results. Other discovery methods from the *CausalLearn library* [1] produced similar graphs. The discovered graph in Figure 2 shows the relationships for the covariates in SUMO.

The same trace data may be used to fit the SCM; however, that involves a pre-processing step to prevent negative values and allow for comparison of distributions.

SAT generation - given a causal graph G, possible SATs are generated as described in section III-A1.

2) Loop execution: Once the three representation spaces are generated, they are continuously updated to be used during the execution of the adaptation loop—see Algorithm 1.

Monitoring takes as input the DRL agent's log traces tagged with the agent's internal timestamps and the policy π_j currently in execution. Time unit j corresponds to the j^{th} full MAPE execution and is different than the agent's time scale. Policy π_j is the outcome of the MAPE execution at $t = t_j$. The monitoring activity outputs traces containing a distribution shift—a perfect detection of the distribution shifts is assumed.

Analysis detects violation of constraint Cs at $t = t_{j+1}$, which triggers the deployment of a *safe policy* and the execution of the full MAPE. The trace produced by policy π_j is then used to *update* the probability distributions in both the causal and symbolic representations, i.e., G and SAT, respectively. These correspond to the marginal expectations (nodes in G), the conditional expectations (edges in G after refitting the SCM), and the covariate distributions for SAT nodes.

Planning loads the updated representation models together with the *causal reasoning engine*—see Figure 4 for inner workings of the causal reasoning engine. The engine (1) selects paths in SAT that best explain the distribution shift, i.e., observation, that caused the constraint violation; (2) for each path, utilizes the SCM and G to generate a set of interventions I; (3) selects the intervention with minimal change in configuration $\bar{\Phi}_{1:N}$ that satisfies the violated constraint to a larger extent.

Execution takes as input the new configurations $\bar{\Phi}_{1:N}$, i.e., the *transferred knowledge* from DRL agent at $t = t_j$, and the policy π_j and uses them as the *head-start* to train the DRL agent for the new policy π_{j+1}. Once the DRL agent is trained, it is deployed, and the adaptation loop terminates.

Knowledge is constantly updated as each of the three representation models changes during the MAPE activities. After the representations are generated, one full MAPE execution is completed, and the subsequent loop starts by monitoring, i.e., collecting traces and identifying constraint violations.

Algorithm 1 Control Loop

1: **Input:** Theory $\mathcal{T}$, Constraints Cs, abduction tree SAT, causal model SCM, agent's Ag current policy π_j
2: **while** true **do**
3:   **if** violated(Cs) **then**
4:     **for all** τ_i in $\mathcal{T}$ **do**
5:       δ = estimate_violation(Cs, τ_i)
6:       $\bar{\Psi}$ = select_causal_paths(SAT, τ_i) . . . . . . Algorithm 2
7:       $\bar{I}$.append (generate_interventions($\bar{\Psi}, G, \delta$)) . . Algorithm 3
8:     **end for**
9:     I^* = min_max($\bar{I}, \mathcal{T}$) . . . . Parsimony condition
10:     Φ = extract_configuration(I^*)
11:     Ag_j = update_models(Φ,π_j)
12:     redeploy_agent(Ag_j)
13:   **end if**
14: **end while**

C. Causal Reasoning Engine

Given a distribution shift on an outcome variable, the causal reasoning engine is concerned with finding the best explanations for it, which involves identifying the violated constraint (line 3 Algorithm 1), selecting causal paths according to theory $\mathcal{T}$ (Algorithm 2), and generating hypothetical interventions I and selecting the best ones based on their explanatory power EP (Algorithm 3).

Generate causal paths - violations of constraints result in instantiations of the corresponding SATs and the distributions of their nodes ($P(V_i)$). After a constraint is violated, an SAT is activated to produce one or more viable intervention paths Ψ, each comprising a list of nodes from the root (constraint) to the leaf (intervention)—see Algorithm 2. Every Ψ from the SAT maps directly to a path in the causal graph G. Line 16 checks the satisfaction of the consistency and entailment conditions—see Section II-C. In turn, each causal path Ψ corresponds to an adjustment set (control variables) that blocks the other parallel paths from the intervention to the outcome variable. These blocked paths can be indirect paths, i.e., going through one or more mediator nodes (see Figure 2 for an example) or *spurious paths*, i.e., backdoor paths, typically harboring a confounder node—see Section II-A.

Discretize intervention space I^{space} - create discrete intervals for the values to be assigned to an intervention covariate. The smaller the intervals, the finer the control over changes in the intervention variable. We set initial values that in future work can be adapted, given the constraints on how sensitive the outcome variable could be to small incremental changes in the interventions.

Deduce hypothetical interventions I - correspond to (1) feasible SAT paths (Algorithm 2), i.e., those that connect the outcome variable to intervention nodes and (2) plausible intervention, i.e., its effect is strong enough to satisfy the violated constraint specified in $\mathcal{T}$.

Generate post-interventional distribution of the outcome variable o_i - in order to compute the effect of an intervention, one also has to (1) instantiate each I_i by selecting one discrete interval from the I^{space}, which is also determined by $\mathcal{T}$ and (2) determine the covariates to control for, which are called adjustment set (Section II-A) and are deduced as listed in Algorithm 2. This procedure is detailed in Algorithm 3.

Compute the explanatory power EP **of hypothetical interventions** I - given an observed outcome $\mathbb{O}^*$ and a set of feasible interventions I_i, we obtain a posterior distribution of the intervention effect δ_i^{pd}.We quantify the explanatory power EP introduced in Section II-C as the rate between changes in the posterior distributions versus the shift on the outcome variable. Essentially, $EP_i \in [0.1, 0.99]$ implies an under-explanation of the shift on $\mathbb{O}^*$ (i.e., I_i would under-correct the violation). Conversely, $EP > 1$ over-explains it (i.e., I_i would be an over-correction). Likewise, $EP_i = 0$ entails no valid explanation, whereas $EP_i = 1$ is a full explanation, i.e., complete overlap of the interventional effect and the observed shift. Assuming that $\Delta(Co) = 11$ is the observed violation expressed as a shift on the number of traffic collisions Co and I_a is the hypothetical intervention that would offset the supposed cause of that violation by training the DRL agent on a decreased friction, e.g., $\Delta(Fr) = 0.5$, ceteris paribus, i.e., desired speed DS remains the same. The explanatory power EP_a is the ratio between (1) the predicted effect $\Delta_{Co}^{pe}(\Delta(Fr) = 0.5)$ (i.e., the difference relative to a baseline before an intervention is applied) and (2) the environment perturbation,i.e., is a shift computed as a difference between the current distribution and the its version before the perturbation. To compute these two differences, we have to fit four models, estimate the cumulative distributions, take their differences, and compute the rate. Finally, we choose the I_i that ranks the highest in EP_i.

Algorithm 2 Select Causal Paths

1: **Input:** $SAT, \mathcal{T}$
2: **Output:** List of all causal paths $\bar{\Psi}$ from the $SAT.root$ to its leaves (intervention variables).
3: $\bar{\Psi}, \Psi = [\ ]$
4: $V = SAT.root$
5: $select\_paths(V, \bar{\Psi}, \Psi)$
6: **return**$\bar{\Psi}$
7: **function** $select\_paths(V, \bar{\Psi}, \Psi)$
8: **if** $V = null$ **then**
9: **return**
10: **end if**
11: Ψ.append(V)
12: **if** V.get_children() is empty **then**
13: $\bar{\Psi}$.append(Ψ)
14: **else**
15: **for all** v in V.get_children() **do**
16: **if** consistent $(v, \mathcal{T})$ **and** $v \cup \mathcal{T} \models V$ **then**
17: $select\_paths(v, \bar{\Psi}, \Psi)$
18: **end if**
19: **end for**
20: **end if**
21: Ψ.pop()

Algorithm 3 Generate Interventions

Input: causal graph G, pathways $\bar{\Psi}$, shift δ.
Output: interventions I_i sorted by explanatory power $EP_i > 0$ of their estimated effects.
$I_i = \bar{\Psi}[i++]$
while $i < \bar{\Psi}.length$ **do**
Ad_i = deduct_adjustment_set(G, I_i)
Ef_i = estimate_effect(I_i, Ad_i)
EP_i = explanatory_power(Ef_i, δ)
if ($EP_i > 0$) **then**
$\Omega = \Omega$.insert_sort(I_i)
end if
end while

The selection of the intervention can be made in various ways, for instance, by Multi-Armed Bandits (to trade off the maximum intervention effect and its uncertainty), Wasserstein distance (to rank the expected effect distributions), or a parsimony criterion (the minimum intervention that maximizes its effect). After the choice among the interventions is generated by Algorithm 3, we can determine both the new environment configuration Φ and the (prior) policy π for transfer.

Find min-max solution is the final step in the causal reasoning process and aims to check the parsimony condition for the plausible explanations—see Line 9 Algorithm 1 and Section II-C. Given a distribution shift, the parsimony condition is concerned with finding the *minimal* intervention set I that is necessary and sufficient to *maximize* the offset of a distribution shift δ on an outcome variable o caused by the DRL agent's Ag under policy π violating a safety constraint Cs, e.g., breaching maximum collision threshold. Our approach is to determine conditions that explain the distribution shift δ. These conditions correspond to the minimal intervention set I^* that would most likely (i.e., present the highest explanatory power EP) update the DRL agent Ag policy π that satisfies all constraints Cs in the theory $\mathcal{T}$—see Algorithm 3.

IV. Evaluation

A. Overview

The application scenario consists of a DRL agent controlling a traffic light with an unprotected right turn condition reproduced in the SUMO simulator—see Section II-D.

Reward Function - the covariates regulating the environment and their relationships are captured in the causal graph (Figure 2). As we optimize for both traffic flow (minimize WT) and safety (minimize Co), we are faced with the a multi-objective optimization problem of first designing a practical reward function R for the traffic light [55], [68]. Instead of approaching the issue through reward design, we enabled causal reasoning while allowing for a more straightforward reward function that represents cumulative delays (in seconds) caused by vehicles waiting due to traffic congestion or collisions.

Goal - the DRL agent updates every 100 time steps to capture delayed action effects, as collisions may occur after the triggering action. Upon collision, a one-minute penalty is added to the reward. Waiting time is the duration (in seconds) during which the vehicle's speed remains below 0.1 m/s. Hence, the reward in seconds is $R = \sum_{t=1}^{100}(WT(t) + 60 \cdot Co(t))$, where WT is the cumulative waiting time until time t and Co counts 1 if there was a collision at time t, otherwise zero. Assume a theory[5] $\mathcal{T}$ where friction Fr and desired speed DS are the only plausible causes of a violation of a specified constraint on Co and/or WT. The causal reasoning involves choosing paths $\bar{\Psi} \in$ SAT (Figure 3 and Algorithm 2) that explain the observed shift δ^{O}_{Fr} or δ^{O}_{DS}.

Consider that traffic light agents can be trained from scratch (brute force) or on top of a previous policy π and environment configuration Φ. The decision between the two alternatives of updating the DRL agent Ag policy π is made by reasoning over the available environment interventions $< Fr, DS >$ instantiated from the causal model G in Figure 2 that satisfy the violated constraints with the minimum cost <cumulative regret>. Figure 6 illustrates the hypothesis regarding the cumulative reward for the DRL agents with and without knowledge transfer.

B. Experimental setup

Training Procedure - we trained baseline DRL agents by taking the Cartesian product of the sets of two desired speeds DS and two frictions Fr to obtain four configurations - $\Phi_{[1:4]} = \{50, 80\} \times \{0.5, 1\}$ combined with a range of other environmental states like density of cars, number of cars waiting, etc. For each configuration, we trained agents using the Actor-Critic algorithm - A2C [47] (known for good performance in adaptive TSC [14], [60]) and selected the policy with both the highest initial starting point and the final cumulative reward. These agents were used as warm starts for training the source agents to be compared against the knowledge-transferred agents, mitigating the unfair impact of random initializations.

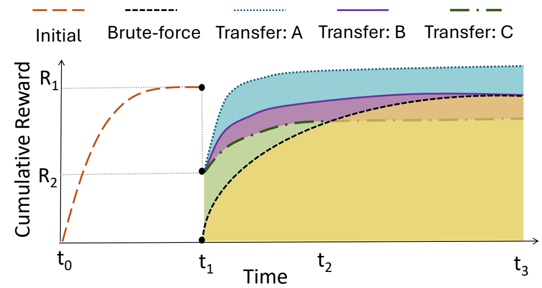


Fig. 6: Illustration of cumulative reward for distinct hypothetical scenarios of knowledge transfer and brute force

Posterior Estimation Procedure - this starts by approximating the Bayesian posterior of the Collision and Waiting Time outcome. This is done both for pre- and post-intervention conditions. For pre-intervention, the observational data are z-score normalized, sampled from a fitted structural causal model (SCM), and de-normalized. We also adopted the Weibull [36] to mitigate data sparsity, avoid negative values, and prevent the violation to the causal positivity assumption.[6] This procedure approximates the Direct D_e and Total T_e effects for interventions on Friction Fr and Desired Speed DS. Then, we compute the Indirect effects I_e by fitting a Weibull distribution on the difference $T_e - D_e$. Concerning the post-intervention, the same function evaluates effects across specified ranges, i.e., I^{spaces} (e.g., $Fr \in [0.5, 1]$ with $DS = 50$ or $Fr = 0.5$ and $Fr \in [50, 80]$), maintaining parameter constraints during sampling and Weibull fitting.

Explanatory Power Quantification - differences between cumulative distribution functions (CDFs) pre- and post-intervention are computed to derive delta CDFs. The ratios of these deltas are calculated to assess the relative impact of each intervention, with results sorted to rank explanatory power. This approach systematically quantifies how interventions on Fr and DS affect the outcomes Co and WT, enabling causal effect comparison. Finally, we compute the differences between the cumulative distribution functions (CDFs) of the pre- and post-intervention outcomes—see Table I.

Explanatory power EP is derived as the ratio between these two differences: $EP = 1 - \frac{|\Delta \mathrm{CDF_{obs}} - \Delta \mathrm{CDF_{pred(i,j)}}|}{\Delta \mathrm{CDF_{obs}}}$, where i and j represent distinct interventions (e.g., Fr vs. DS). Results are then sorted in descending order.

[5]More complex causes can be assumed without changes to approach, e.g., soft interventions on a mediator like the emergency brake (ABS technology) or a mix of cars and trucks.

[6]The proper identification of causal effects requires that control and interventional distributions have common support [32], i.e., every item in the control dataset has non-zero probability in the treatment set.

Observation shifts (ob)			Hypotheses		Predicted effects (pe)		Explanatory power (EP)	
δ^{ob}_{Co}	δ^{ob}_{WT}	Constraint	H_i	$source \xrightarrow{\delta} target$	$\delta^{pe}_{Co}(D_e; T_e)$	$\delta^{pe}_{WT}(D_e; T_e)$	$EP_{Co}(D_e; T_e)$	$EP_{WT}(D_e; T_e)$
0.278	0.972	$Co > 11,\ WT > 38$	$H_{1.A}$	$O_t(50,1) \xrightarrow{\delta} O_{t+1}(50,0.5)$	(-0.201 ; -0.463)	(0.01 ; -0.033)	(0.0 ; 0.0)	(0.01 ; 0.0)
			$H_{1.B}$	$O_t(50,1) \xrightarrow{\delta} O^*_{t+1}(80,1)$	(0.171 ; 0.083)	(0.01 ; 0.024)	**(0.614 ; 0.297)**	**(0.01 ; 0.024)**
-0.381	0.01	$Co > 15,\ WT > 60$	$H_{2.A}$	$O_t(80,0.5) \xrightarrow{\delta} O^*_{t+1}(50,0.5)$	(-0.118 ; -0.105)	(-0.014 ; 0.01)	**(1.689 ; 1.725)**	**(0.0 ; 1.0)**
			$H_{2.B}$	$O_t(80,0.5) \xrightarrow{\delta} O_{t+1}(80,1)$	(0.238 ; 0.434)	(0.01 ; 0.014)	(0.0 ; 0.0)	(1.0 ; 0.581)
0.01	0.01	$Co > 39,\ WT > 42$	$H_{3.A}$	$O_t(50,0.5) \xrightarrow{\delta} O^*_{t+1}(80,0.5)$	(0.01 ; 0.013)	(0.033 ; 0.011)	**(1.0 ; 0.684)**	**(0.0 ; 0.941)**
			$H_{3.B}$	$O_t(50,0.5) \xrightarrow{\delta} O_{t+1}(50,1)$	(0.01 ; 0.014)	(0.01 ; 0.02)	(1.0 ; 0.569)	(1.0 ; 0.031)

TABLE I: The observation shifts $\delta^{ob}_{Co}, \delta^{ob}_{WT}$ affect collision Co and waiting time WT. These shifts violate specified constraints (third column) of an agent (traffic light) trained at time t on an environment and then exposed at time $t+1$ to a modified environment. The cause of these shifts is hypothesized, as for instance, $H_{1.A}$ (decrease in Fr) or $H_{1.B}$ (increase in DS). The relevant predicted effects $\delta^{pe}_{Co}, \delta^{pe}_{WT}$ of these causes were direct (D_e) or total (T_e), whereas indirect effects were consistently zero (hence not shown for visual clarity). The columns $EP_{Co}()$ and $EP_{WT}()$ display the explanatory power of the corresponding effects of the hypothetical intervention H_i on Co, WT. The ground truth of the hypothetical target observed effect is $\mathbb{O}^*$, bold are competing EP within hypotheses H_i, and green cells indicate the correct hypothesis choice based on maximizing EP.

C. Results

RQ1: Trade-off Explanatory Power and Thresholds - we plotted two hypotheses EP across a range of thresholds Figure 7. The right chart shows that for the constraint in the [0.1,30] range, the blue intervention is the only viable explanation (although overshooting). Beyond this range, both blue and orange explain the shift perfectly. The left chart shows a clear choice for the orange intervention for thresholds below 20%, but above 70%, the alternatives produce complete explanations.

RQ2: Effectiveness of Transfer - we prioritized pairs that increase the risk of collisions by either fixing the friction level and increasing speed or vice-versa. This led to the following transfer conditions of source (S) to target (T), i.e., $S_i \rightarrow T_j, i \neq j$:

1) $S_i([50, 80], 1) \rightarrow T_j([50, 80], 0.5)$, keeps speed either 50 or 80, reduces friction to 0.5.
2) $S_i(50, [1, 0.5]) \rightarrow T_j(80, [1, 0.5])$, keeps friction either 1 or 0.5, increases speed to 80.

a) Deployment: We deployed each DRL agent T on an adversarial environment that operates beyond the training configuration ϕ_i, i.e., $DS \in [30, \ldots, 100] \wedge Fr \in [0.25, 1]$.

b) Hypothesized outcomes: Assuming brute force (train without transfer) as a baseline, we hypothesized in Figure 6 that the possible reward curves could show transfer better than brute force (A), indifferent (B), or worse (C).

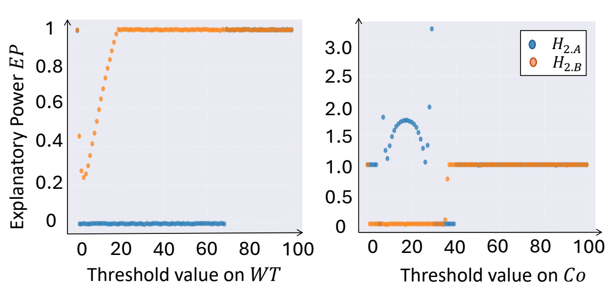


Fig. 7: The choice between hypothetical interventions $H_{2.A}$ and $H_{2.B}$, direct effects on WT (left) and total effects on Co (right), and the behavior of explanatory power over constraint values above the thresholds

Cumulative Regrets Table II presents the results of comparing the trainings by deduced interventions versus brute force, i.e., trained from the baseline DRL agent. Rate λ_C is computed as the mean of the average rate of change of reward, observed over a sliding window with a size of 100,000 steps, applied across the entire diagram. Variance σ_C is calculated as the mean of the variances within the same sliding windows. The reward is the reward of the final episode after a total of $1,000,000$ steps. As seen in Table II, the knowledge-transferred DRL agents consistently outperform the brute force agents in terms of the final reward. For two of the agents, the variance is very high, showing a certain risk of instability when transfer learning, possibly occurring through catastrophic forgetting. In the first row of Table II, the knowledge-transferred agent had a similar learning rate with a much lower variance, showing very stable training (Figure 8). The results in Figure 8 confirm the previous hypothesis illustrated via Figure 6 regarding the cumulative reward with and without knowledge transfer.

Summary of answers to the research questions - (RQ.1) *Is there a trade-off between constraint specification and choices of knowledge transfer?* **Yes**, as in Figure 7 and Table I, explanatory power of hypothetical interventions is sensitive to the threshold specified in the theory $\mathcal{T}$. (RQ.2) *Is the choice of knowledge transfer more effective than the two extremes of training from scratch (zero transfer) or replacing with a pre-trained policy (total transfer)?* **Yes**, in Table II, the brute force training underperformed in all scenarios when evaluated by reward, similarly for the pre-trained policy (Figure 8).

D. Implications

a) Reasoning about transfer operations: The causal representation allows for deciding which knowledge to transfer. Each intervention corresponds to an adjustment set. The difference between the interventions that lead to two DRL agents corresponds to the transfer operations as proposed by [4]. Adding a covariate to the intervention formula consists of closing a path (not reusing the effect communicated through it). Conversely, removing a covariate from the formula opens a path that was closed. Keeping a covariate maintains a path closed (transfers the knowledge about control). If there is a partial shift on a controlled covariate, then one can merge distributions (update knowledge). This allows a cautious adaptation that mitigates side effects from bad controls [15].

b) Reasoning about decision outcomes: The symbolic abduction tree SAT can explain why a chosen posterior was not the best one, for instance, because of partial information distribution shifts. In this sense, we allow both forward reasoning (causal inference, intervention synthesis) and backward reasoning (explainability, self-diagnosis).

c) Complexity: Given the joint probability distribution $\mathscr{P}$ over all n covariates, i.e., nodes in SAT, $P(x_1, \ldots, x_n) = \prod_j P(x_j | x_1, \ldots, x_n)$, storing the data of all conditional probabilities requires a table with d^n or 2^n for binomial variables. However, assuming a *small world structure*[7] underlying the conditional distributions, a graph model is used to discover the structural properties of the probability models—Causal models show small-world structure because tightly linked variables form clusters, with key variables connecting them across short paths. Consequently, we could assume that $\mathscr{P}$ is Markovian, i.e., $\mathscr{P} = \prod_j P(x_j | parents(x_i))$, where $parents(x_i)$ is the set of covariates that are causes of x_i, then complexity decreases from exponential in n to linear, i.e., $O(n2^k)$, where k is the maximum number of parents in the graph (SAT)[8].

d) Efficiency and Scalability: Because the complexity of adapting the SAT is bounded and the causal graph G is generated only once with a complete world representation of all configuration options, the cost of updating these models is predictable. That is not the case with the DRL agent. Hence, decisions to update models via knowledge transfer should take into account these differential costs, particularly because stricter constraints impose more frequent model updates. Our results can serve as a baseline for approaches that freeze policy neural network layers [30] and for representations that explain transfer changes to the neural network [64].

[7]A type of network where most nodes are not neighbors but can be reached from every other by a small number of steps, due to high clustering and short average path lengths.

[8]Note that because nodes can be connected in many different ways, the number of possible distinct graphs with the same number of nodes can grow very quickly [59], e.g., three nodes can be arranged in 25 different acyclic graphs (DAG), 5 nodes 29281 DAGs, and 7 to 1.1E+8 DAGs.

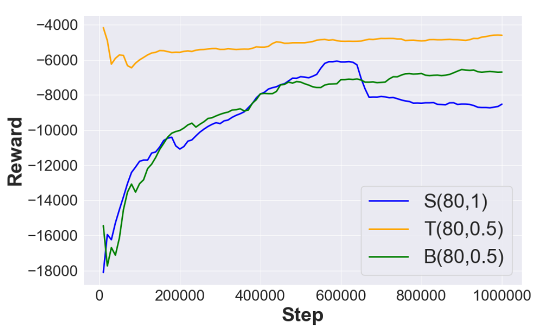


Fig. 8: Brute force (Green), Transfer (Yellow) is a warm-start from pre-trained policy π_{t-1} (Blue) fine-tuned on the intervention with the highest explanatory power.

E. Threats to Validity

External validity: one could increase the efficiency of training by adapting the reward function (reward design [55]). Instead, we opted to first determine the best-known configuration for training. Therefore, we cannot guarantee that our approach would generalize if one were to simultaneously adapt the reward function. Future work should investigate the generalizability of more complex interventions and system domains (e.g., robotics, cloud services, etc.). *Internal validity*: During training from transfer, certain DRL agents showed erratic cumulative reward behavior (e.g., collapse and later recovery). Because the training instability happens in later steps, the threat of policy collapse [18] must be monitored. Another threat is the risk of an underspecified causal model, which can result from hidden confounding (lack of causal sufficiency) or selection bias (incomplete simulation traces). We mitigated the former by relying on domain knowledge to determine which edges are not possible, whereas for the latter, we cover a grid of configuration values, hence guaranteeing a homogeneous coverage of the state-space. *Construct validity*: we chose convergence metrics that capture both the speed and uncertainty about the training relative to a baseline DRL agent. While these make sense when one is focused on efficiency, these might not reflect the efficacy of convergence, e.g., robustness and stability. For those, one should consider metrics like average and discounted rewards [17].

V. State-of-the-Art

A. Transfer Learning for SAS

ML-based systems, including SAS, are susceptible to concept drift, which diminishes the accuracy of prediction models [27]. Several studies explore concept drift detection [26], model mispredictions [11], and the emergence of new knowledge [25], capturing that knowledge for explaining the effects of adaptations [19], and, hence, enabling reasoning about this knowledge transfer [4]. We complement these approaches by interleaving causal reasoning and self-adaptation.

Transfer condition	rate λ_C	variance σ_C	reward
$Warmup \rightarrow B(80, 0.5)$	**9.51e-03**	2.65e+05	-6694
$S(80, 1) \rightarrow T(80, 0.5)$	1.16e-03	**1.75e+04**	**-4599**
$Warmup \rightarrow B(50, 0.5)$	7.63e-03	**3.03e+05**	-8513
$S(50, 1) \rightarrow T(50, 0.5)$	**5.54e-01**	3.58e+09	**-5795**
$Warmup \rightarrow B(80, 1)$	6.55e-03	**2.75e+05**	-8524
$S(50, 1) \rightarrow T(80, 1)$	**2.66e+00**	5.38e+10	**-6839**
$Warmup \rightarrow B(80, 0.5)$	**9.51e-03**	2.65e+05	-6694
$S(50, 0.5) \rightarrow T(80, 0.5)$	3.29e-04	**6.87e+03**	**-5291**

TABLE II: Convergence metrics of training via brute force versus knowledge transferred DRL agents. *Warmup* stands for a pre-initialized DRL agent, while "B" brute force, "S" source, and "T" target DRL agents are trained on a given $< DS, Fr >$ configuration. The symbol $\rightarrow$ signifies that the right-hand-side agent was initialized with the policy on the left before training.

Concerning reasoning about knowledge transfer, rules can be precisely specified as if-then-else statements, which can speed up the learning of conditions for action application, e.g., thresholds [9] and specify safety constraints. However, logic-based reuse of learned rules is still vulnerable to misspecification, which we mitigate via knowledge transfer.

Reusing prior knowledge may not be adequate to address the large adaptation spaces or the associated costs and latency of planning adaptation. Quin et al. [53] employ classification and regression to narrow the adaptation space. While this approach is scalable, it remains uncertain how verification and validation of transfer perform under the strict planning time constraints.

In RL-based SAS, the goal is to train agents to act autonomously. We go beyond that by endowing the agent with autonomy on how it chooses to learn. Our approach splits the adaptation goal into (1) understanding the observable associations within the environment (causal structure) and (2) understanding the corresponding adaptation choices (decision structure). While the causal associations are obtained by measuring the effect of interventions on the environmental configurations, the adaptation choices correspond to the paths through these associations. Conversely, in current approaches [16], the environment causal associations are implicitly captured in the MDPs and the learned policies, which prevents reasoning about knowledge transfer trade-offs.

B. Safe RL

Among the various safe DRL techniques [22], three techniques are close to our approach: *reward shaping*, *shielding*, and *safe exploration*. *Reward shaping* methods encode safety constraints in a loss function [6]. *Shielding* [10], [56] utilizes a logical safety specification to prevent the system from taking potentially unsafe actions. *Safe exploration* methods restrict the training process to learn only safe actions [29].

Symbolic logic can be combined with DRL [57] for safe autonomous driving. Safety constraints are captured via a set of FOL rules that define the safe action space. et al. [62] consider an MDP where the unsafe actions are avoided. The safety constraints are modeled as a Gaussian process that allows the algorithm to reason about state-action pair safety.

Contrary to these approaches, we do not require the specification of constraints at the lower level of DRL agent actions. Instead, we specify constraints as the factors (covariates) that influence safety and performance goals. This is more appropriate when reasoning about policies that have delayed and diffuse effects on rewards.

C. Meta-RL

A promising approach to accelerate the standard RL-based solutions for SAS is Meta-RL. Existing approaches are mainly divided into *gradient-based* [20] and *memory-based* adaptations. In the former, meta-learning is done by back-propagation through the inner-loop update [67]. However, in the latter, policy updates can usually only happen after entire batches of data have been collected, whereas for *online adaptation*, the policy needs to incorporate new information and update the policy after every timestep, e.g., as soon as a concept drift is detected. Memory-based approaches, on the other hand, use Recurrent Neural Networks (RNNs) for Meta-Learning [46] where the inner workings of the RNN, i.e., the updating of the hidden state, is a learning algorithm itself. In RNN-based approaches for Meta-RL, the higher-level learning (loop) remains a black-box, hence hindering explainability and reasoning over the DRL agent's behavior [70]. Our approach distinguishes from meta-RL by enabling explicit causal explainability.

D. Integrating Symbolic Planning and RL

Recent research has sought to merge symbolic planning with DRL to improve performance and data efficiency, e.g., [38], [44], [65]. These approaches aim to integrate a high-level planner that suggests sub-goals to be executed by a low-level DRL model, thus relying on pre-defined environment dynamics, such as high-level action schema with pre- and post-conditions. Our approach differs from symbolic automated planning-based methods as it is purely an RL approach, i.e., it does not have access to a handcrafted action schema or a reward function. Instead, we learn the policy from past experiences and knowledge transfer.

E. Causal Reasoning for SAS

Jaimini et al. [33] propose a causal NeSy vision for smart manufacturing where they use causal discovery methods to build a causal representation as a Knowledge Graph (KG). Causal representations were also shown to improve policy learning in single RL agents [8] and knowledge transfer in multi-agent RL scenarios [40]. Comparatively, we combine causal representation and abductive reasoning to improve policy learning via knowledge transfer. Similarly to [66], we contribute to filling the known gap [54] of employing causal reasoning as a means to endow autonomous systems with adaptive behavior, which is particularly challenging to achieve in non-stationary environment conditions [26], [53]. Ultimately, by enabling causal reasoning within the MAPE-K loop of self-adaptive DRL agents, we are combining neuro-symbolic AI [7] and causal AI [49] methods with the vision of improving training costs and more cautious (safer) system adaptations [4].

VI. Conclusion

We presented a framework for neuro-symbolic self-adaptation that enables causal reasoning for learning-enabled safety-critical systems. Our empirical evaluation answers the research questions positively (1) there is a trade-off between threshold setting and choices of transfer, and (2) a trade-off between the extremes of transfer (zero or complete). Concerning knowledge learning, in future work, we aim to integrate uncertainty into the explanatory power measures, allow for partial mediation, and non-stationary processes with continuous shifts. On the symbolic direction, we plan to contemplate more complex constraints on path traversing and endow the current framework with meta-self-awareness, which would be an extension of our ongoing work on causal knowledge transfer in non-stationary multi-agent RL settings [40].

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